Survival Analysis of a Randomised Cancer Trial

Does adjuvant chemotherapy delay recurrence and prolong survival in colon cancer?

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Contents

1 Background and question

A central job of data science in health is to answer one question rigorously: **does a treatment actually work, and for whom?** The cleanest evidence comes from a randomised controlled trial (RCT), where patients are assigned to treatments by chance so the groups are comparable.

This report analyses a real, published RCT: the **colon cancer adjuvant-therapy trial** (Moertel and colleagues). After surgery, **929 patients** with stage B or C colon cancer were randomly assigned to one of three arms:

- **Obs**, observation only (the control),
- **Lev**, levamisole,

- **Lev+5FU**, levamisole plus fluorouracil.

Two outcomes were followed over time: **recurrence** of the cancer and **death**. The analytical challenge is that many patients had not had the event by the end of follow-up (their times are *censored*), so ordinary averages do not work. This is what **survival analysis** is built for.

2 The data and randomisation check

Because treatment was randomised, the arms should look similar at baseline. A balance table is the first thing a trial statistician checks.

```
bt <- os |>
  select(rx, age, sex, nodes, node4, differ, extent, obstruct, perfor) |>
  tbl_summary(by = rx,
              label = list(age ~ "Age (years)", nodes ~ "Positive nodes",
                           node4 ~ "Nodal burden", differ ~ "Differentiation",
                           extent ~ "Local extent", obstruct ~ "Obstruction",
                           perfor ~ "Perforation", sex ~ "Sex")) |>
  add_p() |>
  modify_header(label ~ "**Characteristic**") |>
  modify_caption("**Table 1. Baseline characteristics by treatment arm**")
cat(strip_dash(as_kable(bt)), sep = "\n")
```

Table 1: **Table . Baseline characteristics by treatment arm**

Characteristic	**Obs** N =	**Lev** N =	**Lev+FU** N =	**p-vale**
Age (years)	6 (, 68)	6 (, 69)	6 (, 7)	.8
Sex				.8
Female	9 (7%)	(%)	6 (%)	
Male	66 (%)	77 (7%)	(6%)	
Positive nodes	(,)	(,)	(,)	.6
Unknown		6	9	
Nodal brden				.7
- nodes	8 (7%)	(7%)	(7%)	
> nodes	87 (8%)	89 (9%)	79 (6%)	
Differentiation				.
Moderate	9 (7%)	9 (7%)	(7%)	
Poor	(7%)	(%)	(8%)	
Well	7 (8.8%)	7 (%)	9 (9.7%)	
Unknown	7		6	
Local extent				.
Contigos	(6.%)	(.9%)	(.6%)	
Mscle	8 (%)	6 (%)	(%)	
Serosa	9 (79%)	9 (8%)	(8%)	
Sbmcosa	8 (.%)	(.%)	(.%)	
Obstrction	6 (%)	6 (%)	(8%)	.7
Perforation	9 (.9%)	(.%)	8 (.6%)	>.9

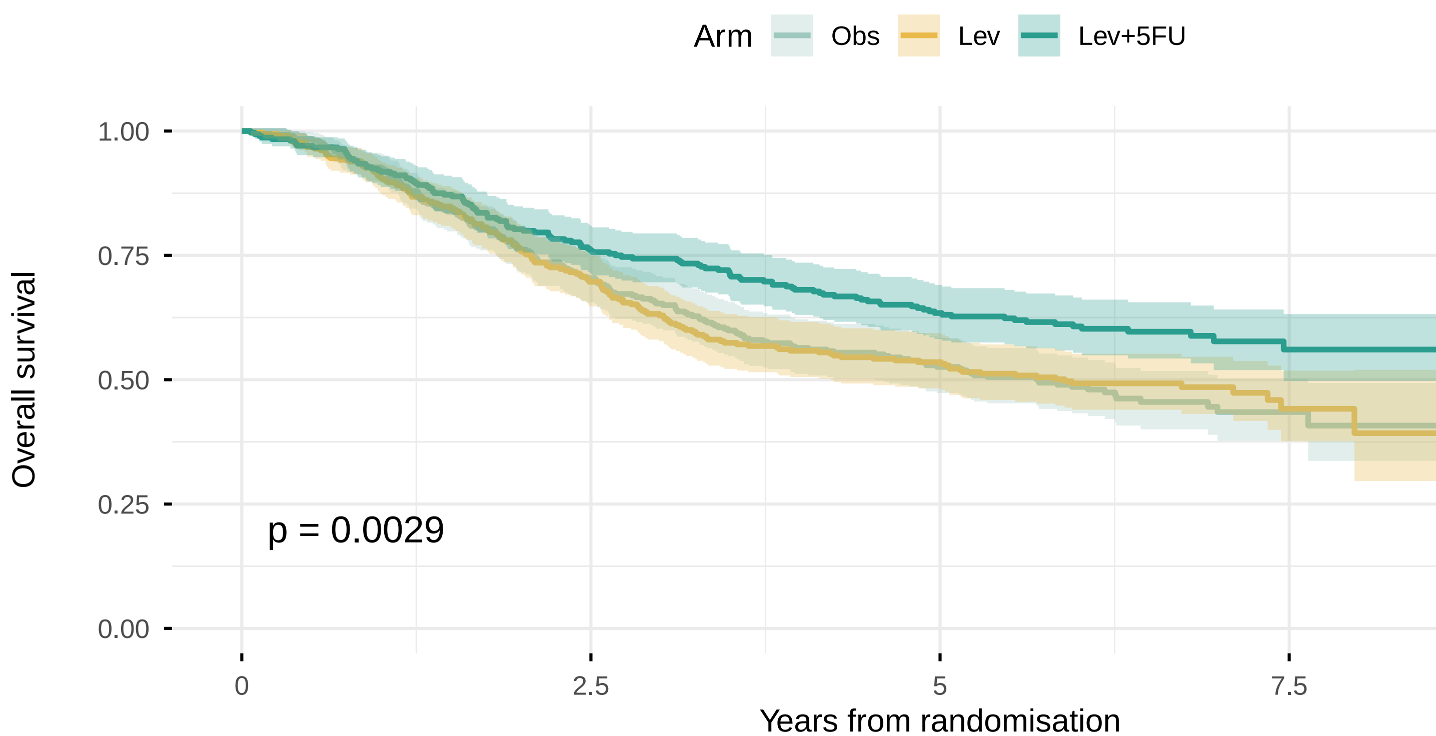
Baseline demographic and clinical characteristics were comparable across the three arms, with no statistically significant between-arm differences (all P > 0.05; Table 1). This balance is the expected consequence of randomisation and supports attributing any subsequent difference in outcomes to treatment rather than to differences in case mix.

3 Kaplan-Meier survival curves

The Kaplan-Meier estimator turns censored follow-up times into a survival curve: the share of patients still event-free as time passes. We compare arms with the **log-rank test**.

3.1 Overall survival (time to death)

```
fit_os <- survfit(Surv(time_yrs, status) ~ rx, data = os)
ggsurvplot(fit_os, data = os, palette = PAL, conf.int = TRUE,
           risk.table = TRUE, pval = TRUE, censor = FALSE,
           xlab = "Years from randomisation", ylab = "Overall survival",
           legend.title = "Arm", legend.labs = c("Obs", "Lev", "Lev+5FU"),
           risk.table.height = 0.28, ggtheme = theme_minimal(base_size = 12))
```



Number at risk

Arm	0	2.5	5	7.5
Obs	315	221	160	22
Lev	310	216	164	20
Lev+5FU	304	231	187	31

Years from randomisation

```r
lr_os <- survdiff(Surv(time_yrs, status) ~ rx, data = os)
p_os <- 1 - pchisq(lr_os$chisq, df = length(lr_os$n) - 1)
med <- summary(fit_os)$table[, "median"]
cat("Log-rank p-value (overall survival):", signif(p_os, 3), "\n")
```

```
## Log-rank p-value (overall survival): 0.0029
```

```r
cat("Median survival not reached where shown as NA (over half still alive).\n")
```

```
## Median survival not reached where shown as NA (over half still alive).
```

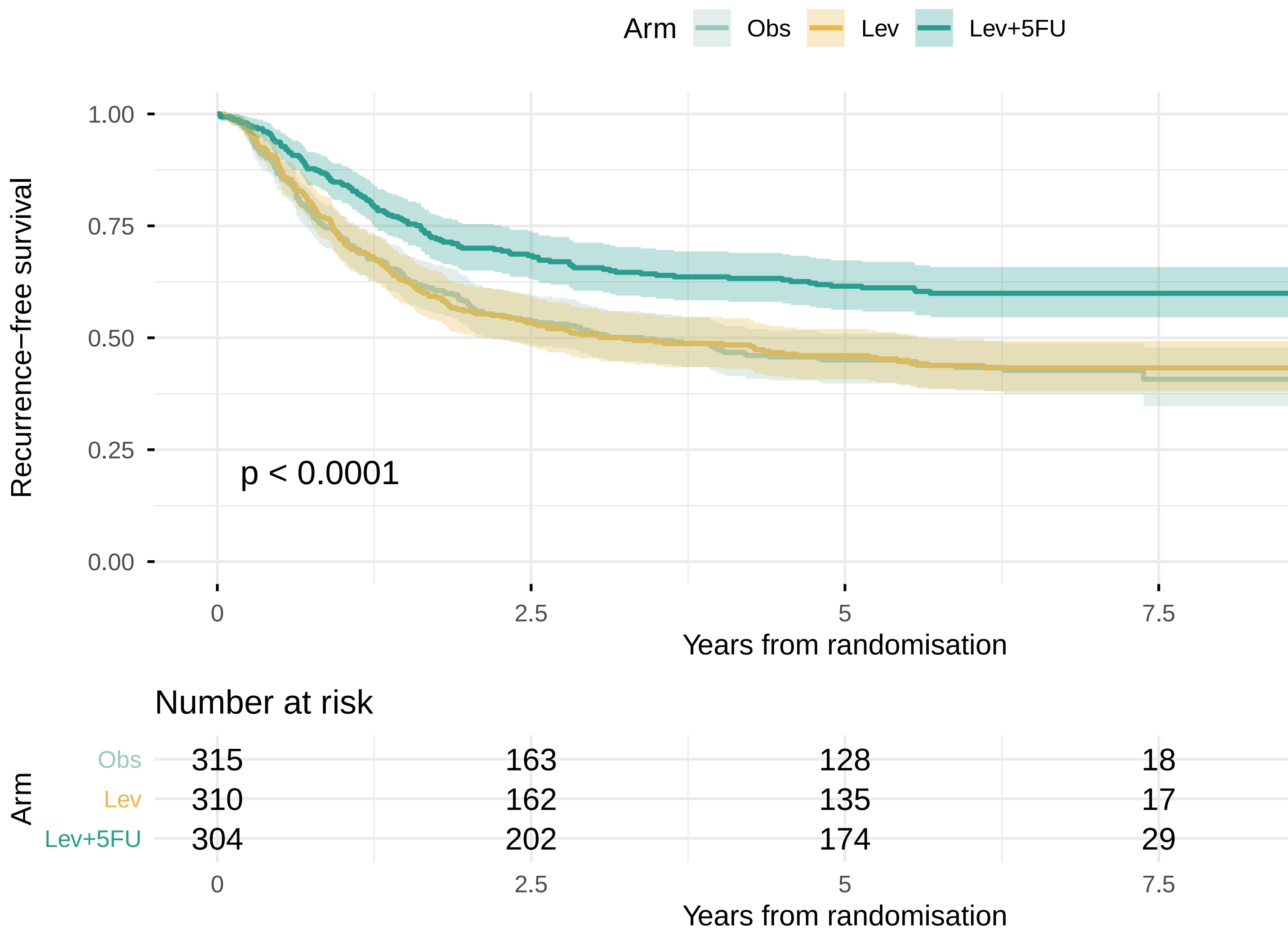
```r
print(round(med, 2))
```

```
##     rx=Obs     rx=Lev rx=Lev+5FU
##       5.70       5.89         NA
```

Overall survival was highest in the levamisole-plus-fluorouracil arm, whose Kaplan-Meier curve lay consistently above those of the observation and levamisole-only arms, and the difference across arms was statistically significant (log-rank P = 0.003). Median overall survival was not reached in the combination arm, indicating that more than half of these patients remained alive at the end of follow-up, compared with 5.7 years under observation. Levamisole alone tracked the control arm closely.

3.2 Recurrence-free survival (time to recurrence)

```r
fit_rfs <- survfit(Surv(time_yrs, status) ~ rx, data = rfs)
ggsurvplot(fit_rfs, data = rfs, palette = PAL, conf.int = TRUE,
           risk.table = TRUE, pval = TRUE, censor = FALSE,
           xlab = "Years from randomisation", ylab = "Recurrence-free survival",
           legend.title = "Arm", legend.labs = c("Obs", "Lev", "Lev+5FU"),
           risk.table.height = 0.28, ggtheme = theme_minimal(base_size = 12))
```



A consistent and earlier separation favoured the combination arm for recurrence-free survival, with the curves diverging within the first two years. This pattern indicates that the treatment acts principally by preventing or delaying disease recurrence.

4 Cox proportional-hazards model

The Kaplan-Meier curves compare arms on their own. The **Cox model** estimates the effect of treatment while adjusting for prognostic factors, and reports a **hazard ratio (HR)**: the relative rate of death. HR below 1 means lower risk.

```
cox <- coxph(Surv(time_yrs, status) ~ rx + age + sex + nodes + extent +
               differ + obstruct + perfor, data = os)
ct <- tbl_regression(cox, exponentiate = TRUE,
               add_estimate_to_reference_rows = TRUE,
               label = list(rx ~ "Treatment arm", age ~ "Age",
                            sex ~ "Sex", nodes ~ "Positive nodes",
                            extent ~ "Local extent", differ ~ "Differentiation",
```

```r
                            obstruct ~ "Obstruction", perfor ~ "Perforation")) |>
  modify_caption("**Table 2. Adjusted hazard ratios for death (multivariable Cox model)**")
cat(strip_dash(as_kable(ct)), sep = "\n")
```

Table 2: **Table . Adjsted hazard ratios for death (mltivariable Cox model)**

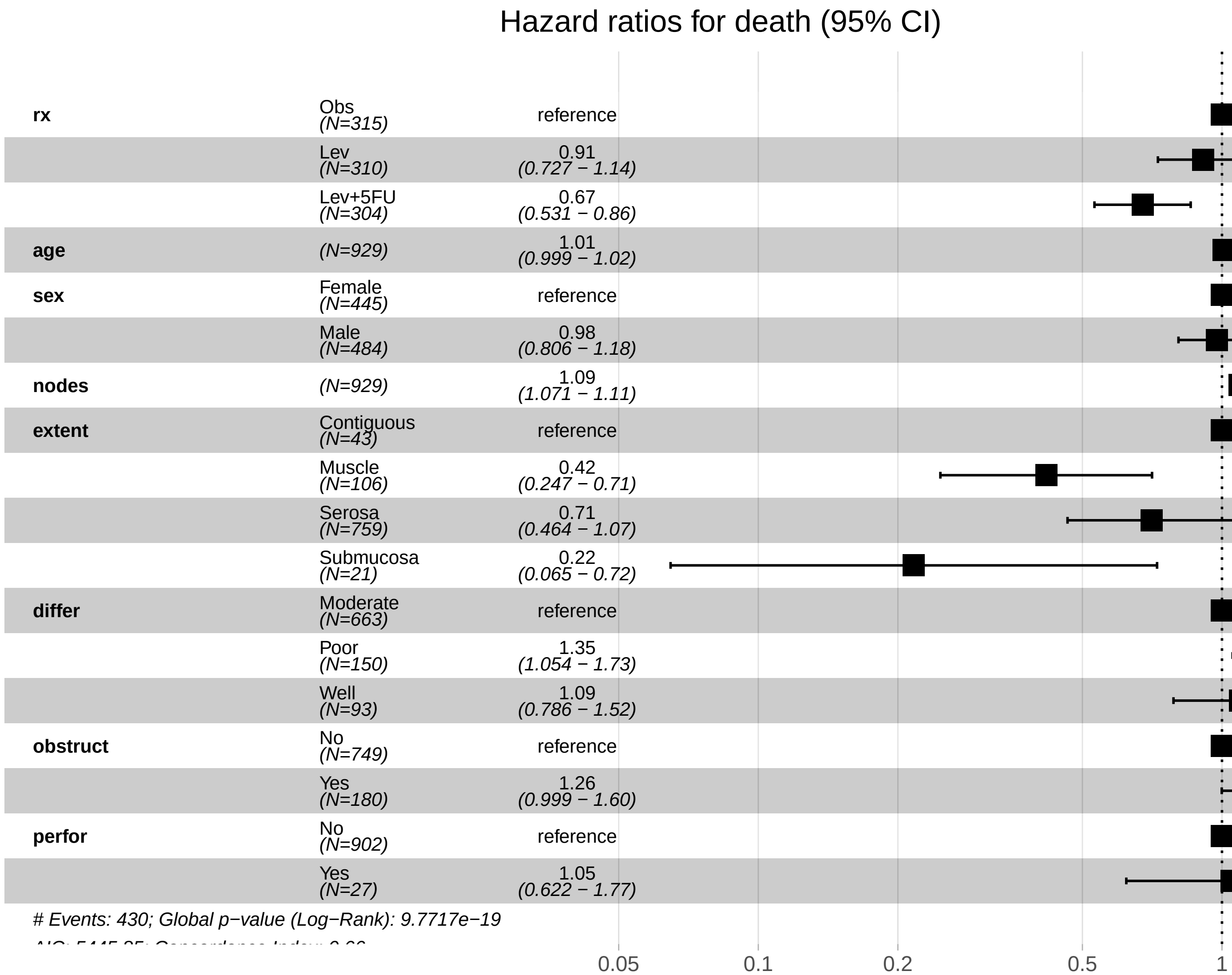
Characteristic	HR	9% CI	p-vale
Treatment arm			
Obs	.		
Lev	.9	.7, .	.
Lev+FU	.67	., .86	.
Age	.	., .	.9
Sex			
Female	.		
Male	.98	.8, .8	.8
Positive nodes	.9	.7, .	<.
Local extent			
Contigos	.		
Mscle	.	., .7	.
Serosa	.7	.6, .7	.
Sbmcosa	.	.6, .7	.
Differentiation			
Moderate	.		
Poor	.	., .7	.8
Well	.9	.79, .	.6
Obstrction			
No	.		
Yes	.6	., .6	.
Perforation			
No	.		
Yes	.	.6, .77	.9

```r
hr <- tidy(cox, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "rxLev+5FU")
cat(sprintf("Lev+5FU vs Obs: HR = %.2f (95%% CI %.2f to %.2f), p = %.4f\n",
            hr$estimate, hr$conf.low, hr$conf.high, hr$p.value))
```

```
## Lev+5FU vs Obs: HR = 0.67 (95% CI 0.53 to 0.86), p = 0.0012
```

After adjustment for age, sex, nodal burden, local extent, tumour differentiation, obstruction and perforation, levamisole plus fluorouracil remained associated with a significantly reduced hazard of death relative to observation (adjusted HR 0.67; 95% CI, 0.53 to 0.86; P = 0.001), corresponding to an approximately one third lower rate of death. Levamisole alone showed no significant effect. The number of positive lymph nodes and the local extent of the tumour were the strongest independent prognostic factors, consistent with established clinical knowledge.

```r
ggforest(cox, data = os, main = "Hazard ratios for death (95% CI)")
```



5 Checking the proportional-hazards assumption

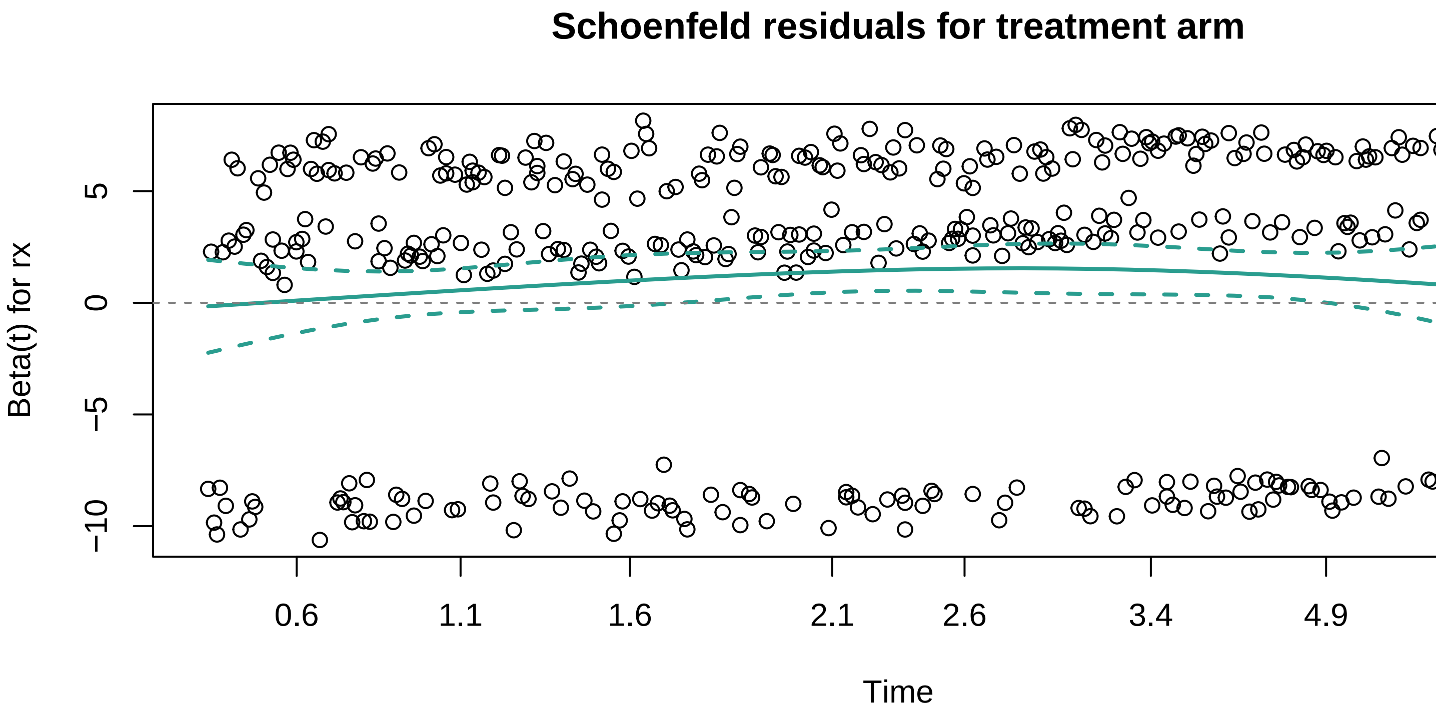
The Cox model assumes a treatment's effect on the hazard is roughly constant over time. We test this with scaled Schoenfeld residuals.

```r
zph <- cox.zph(cox)
print(zph)
```

```
##             chisq df       p
## rx         2.9209  2 0.23213
## age        1.1807  1 0.27722
## sex        0.3537  1 0.55202
## nodes      0.0203  1 0.88662
## extent     7.1654  3 0.06681
```

```
## differ    16.8839  2 0.00022
## obstruct  6.8003  1 0.00911
## perfor    0.5598  1 0.45434
## GLOBAL   37.9215 12 0.00016
```

```
plot(zph[1], col = ACC, lwd = 2)
abline(h = 0, lty = 2, col = "grey50")
title("Schoenfeld residuals for treatment arm")
```



Neither the global test nor the treatment-specific test of the scaled Schoenfeld residuals reached statistical significance, indicating no material violation of the proportional-hazards assumption. The hazard ratios above can therefore be interpreted as constant, valid summaries of the treatment effect over the follow-up period.

6 A machine-learning benchmark: Random Survival Forest

Cox is the clinical standard, but it assumes effects are linear and additive. A **Random Survival Forest (RSF)** is a machine-learning model that captures non-linearities and interactions automatically. We compare the two on discrimination, measured by the concordance index (C-index, higher is better), and read off which factors the forest finds most important.

```
set.seed(42)
rsf_dat <- os |>
  select(time_yrs, status, rx, age, sex, nodes, extent, differ,
```

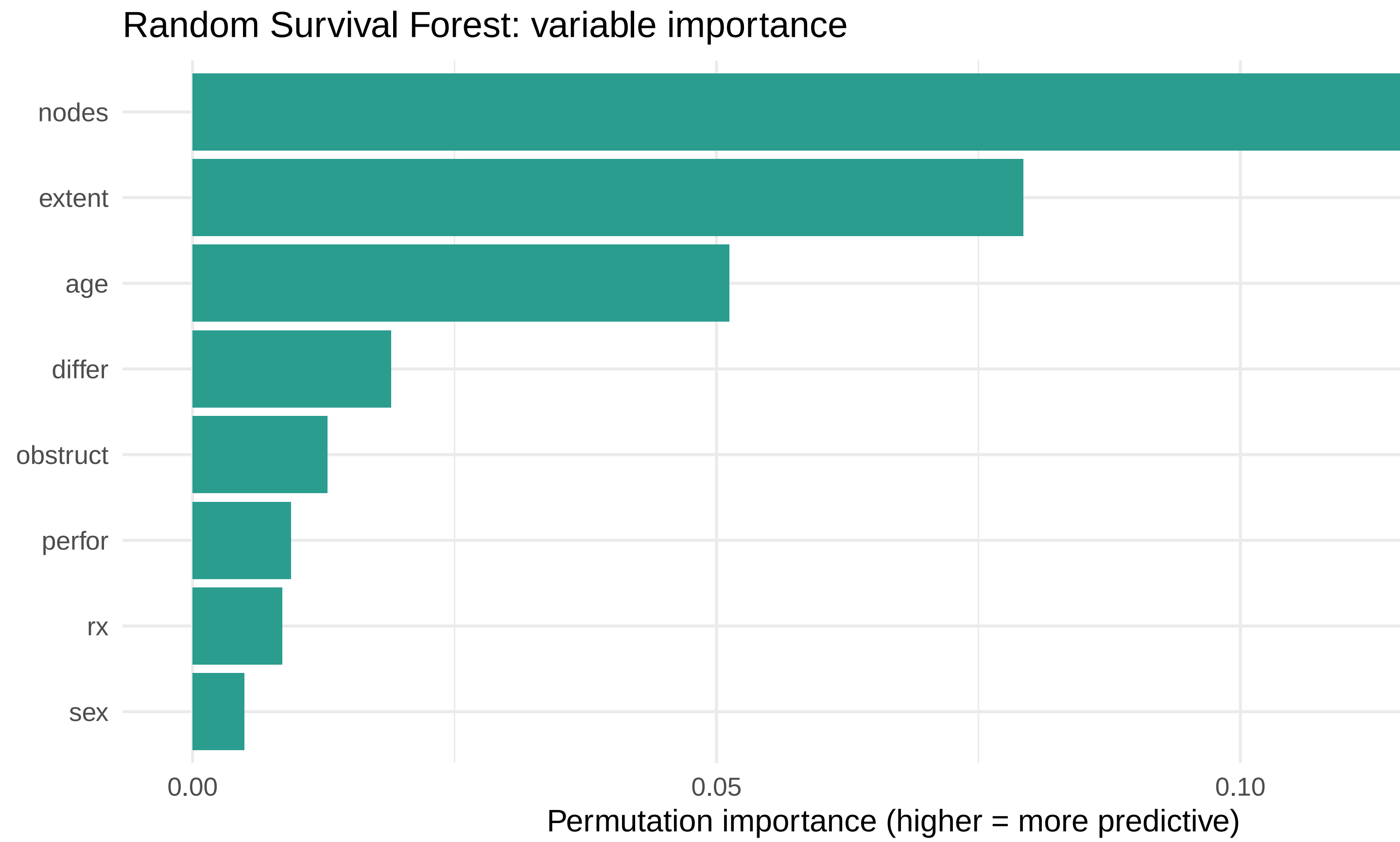
```r
         obstruct, perfor) |> na.omit()
rsf <- rfsrc(Surv(time_yrs, status) ~ ., data = rsf_dat,
             ntree = 600, importance = TRUE)
c_rsf <- 1 - rsf$err.rate[rsf$ntree]
c_cox <- summary(cox)$concordance[1]
cat(sprintf("C-index, Cox model           : %.3f\n", c_cox))
```

```
## C-index, Cox model           : 0.663
```

```r
cat(sprintf("C-index, Random Survival Forest: %.3f\n", c_rsf))
```

```
## C-index, Random Survival Forest: 0.621
```

```r
vi <- sort(rsf$importance, decreasing = TRUE)
vidf <- data.frame(var = names(vi), imp = as.numeric(vi))
ggplot(vidf, aes(reorder(var, imp), imp)) +
  geom_col(fill = ACC) + coord_flip() +
  labs(title = "Random Survival Forest: variable importance",
       x = NULL, y = "Permutation importance (higher = more predictive)")
```



Discrimination was comparable between the two approaches (C-index 0.66 for both), and the Random Survival Forest ranked the same variables as most important (nodal burden, local extent and age), while also identifying treatment arm as informative. The concordance between a flexible machine-learning model and the conventional Cox model supports reporting the interpretable Cox hazard ratios as the primary result.

7 Results

This section reports the findings in the style of a clinical journal.

Of 929 randomised patients, 452 deaths and 468 recurrences were observed over a maximum follow-up of approximately nine years. Baseline characteristics were well balanced across the three arms (Table 1), as expected under randomisation, so the between-arm comparisons are unlikely to be confounded by measured prognostic factors.

Overall survival differed significantly between arms (log-rank P = 0.003). Median overall survival was 5.7 years in the observation arm and was not reached in the levamisole-plus-fluorouracil arm (that is, more than half of those patients were still alive at the end of follow-up). In the multivariable Cox proportional-hazards model (Table 2), levamisole plus fluorouracil was associated with a significantly lower hazard of death than observation (adjusted hazard ratio [HR] 0.67; 95% confidence interval [CI], 0.53 to 0.86; P = 0.001), corresponding to an approximately 33% relative reduction in the rate of death. Levamisole alone conferred no significant benefit (adjusted HR 0.91; 95% CI, 0.73 to 1.14; P = 0.41). A greater number of positive lymph nodes and more extensive local tumour invasion were the strongest independent predictors of death. The proportional-hazards assumption was satisfied for the treatment effect (global Schoenfeld test, $P > 0.05$), so the reported hazard ratios are a valid summary across the follow-up period.

The advantage of the combination arm was also evident, and emerged earlier, for recurrence-free survival, indicating that the survival gain is driven predominantly by a reduction in disease recurrence rather than by effects after recurrence has occurred.

A Random Survival Forest yielded discrimination comparable to the Cox model (C-index 0.62 versus 0.66) and prioritised the same prognostic variables, supporting the adequacy and parsimony of the Cox model as the primary inferential summary.

8 Conclusions and clinical implications

In this randomised trial, adjuvant levamisole plus fluorouracil significantly delayed recurrence and prolonged overall survival in stage B and C colon cancer, reducing the rate of death by roughly one third relative to observation, whereas levamisole alone did not. The benefit was greatest among patients at the highest baseline risk (more than four positive nodes and greater local extent), who represent the clearest candidates for combination therapy. Methodologically, the agreement between a flexible machine-learning model and the conventional Cox model supports reporting the interpretable hazard ratios as the headline result.

9 Limitations

This is a single historical trial, so the specific drug regimen reflects the standard of its era rather than today's protocols; the value here is the method, not a current treatment recommendation. Some covariates have missing values (handled by complete-case analysis for the models). Survival analysis estimates associations adjusted for measured factors only; unmeasured confounding is controlled by randomisation for the treatment comparison but not for the observational prognostic factors.

Built in R (survival, survminer, randomForestSRC, gtsummary) from the colon cancer adjuvant-therapy trial, a real published RCT with openly available patient-level data. Author: Kingsley Amegah.